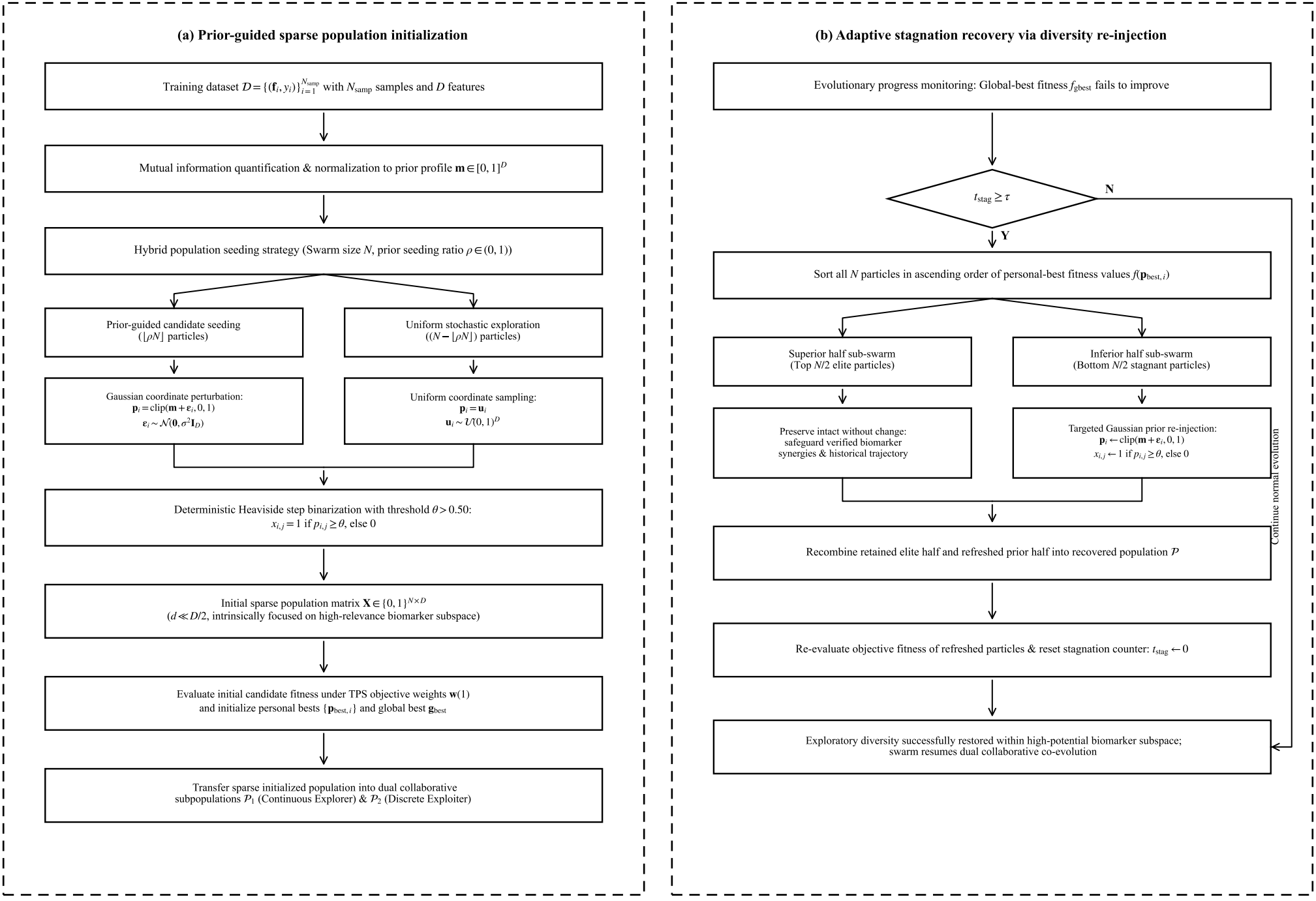

(a) Prior-guided sparse population initialization
Training dataset $\mathcal{D}=\{(\mathbf{f}_i, y_i)\}_{i=1}^{N_{\text{samp}}}$ with N_{samp} samples and D features
Mutual information quantification & normalization to prior profile $\mathbf{m}\in[0,1]^D$
Hybrid population seeding strategy (Swarm size N, prior seeding ratio $\rho\in(0,1)$)
Prior-guided candidate seeding
($\lfloor\rho N\rfloor$ particles)
Uniform stochastic exploration
($(N-\lfloor\rho N\rfloor)$ particles)
Gaussian coordinate perturbation:
$\mathbf{p}_i=\text{clip}(\mathbf{m}+\boldsymbol{\varepsilon}_i, 0, 1)$
$\boldsymbol{\varepsilon}_i\sim\mathcal{N}(\mathbf{0},\sigma^2\mathbf{I}_D)$
Uniform coordinate sampling:
$\mathbf{p}_i=\mathbf{u}_i$
$\mathbf{u}_i\sim\mathcal{U}(0,1)^D$
Deterministic Heaviside step binarization with threshold $\theta>0.50$:
$x_{i,j}=1$ if $p_{i,j}\geq\theta$, else 0
Initial sparse population matrix $\mathbf{X}\in\{0,1\}^{N\times D}$
($d\ll D/2$, intrinsically focused on high-relevance biomarker subspace)
Evaluate initial candidate fitness under TPS objective weights $\mathbf{w}(1)$
and initialize personal bests $\{\mathbf{p}_{\text{best},i}\}$ and global best $\mathbf{g}_{\text{best}}$
Transfer sparse initialized population into dual collaborative
subpopulations $\mathcal{P}_1$ (Continuous Explorer) & $\mathcal{P}_2$ (Discrete Exploiter)
(b) Adaptive stagnation recovery via diversity re-injection
Evolutionary progress monitoring: Global-best fitness f_{gbest} fails to improve
$t_{\text{stag}}\geq\tau$
N
Y
Sort all N particles in ascending order of personal-best fitness values $f(\mathbf{p}_{\text{best},i})$
Superior half sub-swarm
(Top $N/2$ elite particles)
Inferior half sub-swarm
(Bottom $N/2$ stagnant particles)
Preserve intact without change:
safeguard verified biomarker
synergies & historical trajectory
Targeted Gaussian prior re-injection:
$\mathbf{p}_i\leftarrow\text{clip}(\mathbf{m}+\boldsymbol{\varepsilon}_i, 0, 1)$
$x_{i,j}\leftarrow 1$ if $p_{i,j}\geq\theta$, else 0
Continue normal evolution
Recombine retained elite half and refreshed prior half into recovered population $\mathcal{P}$
Re-evaluate objective fitness of refreshed particles & reset stagnation counter: $t_{\text{stag}}\leftarrow 0$
Exploratory diversity successfully restored within high-potential biomarker subspace;
swarm resumes dual collaborative co-evolution